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BAHÇEŞEHİR UNIVERSITY

FACULTY OF ENGINEERING AND NATURAL SCIENCES

CAPSTONE FINAL REPORT

Relationship Between Electricity Market Clearing Price (PTF) and Market Value of Energy Companies: The Case of AKSEN

Burak Can Ağgül-Management Engineering

İlker Eser-Energy Systems Engineering

Melike Koygun-Management Engineering

Utku Süleymanoğlu-Management Engineering

Barış Egemen Tokul-Energy Systems Engineering

Assist. Prof. Mehmet Emin Yıldız-Management Engineering

Assist. Prof. Hüseyin Günhan Özcan-Energy Systems Engineering

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STUDENT DECLARATION

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ABSTRACT

Relationship Between Electricity Market Clearing Price (PTF) and Market Value of Energy Companies: The Case of AKSEN

Burak Can Ağgül, İlker Eser, Melike Koygun, Utku Süleymanoğlu, Barış Egemen Tokul

Faculty of Engineering and Natural Sciences

Advisor: Asst. Prof. Mehmet Emin Yıldız
Asst. Prof. Hüseyin Günhan Özcan

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In this research, as Management Engineers and Energy Systems Engineers, we examined whether fluctuations in the energy market directly caused fluctuations in the stock exchange, in this case, the electricity index of Borsa İstanbul, with different methodologies and correlations to historic evidence of different commodities. We also explored how electricity is priced in Turkish markets and its different methodologies, mainly Market Clearing Price (PTF) and Merit Order. In all our works, EPİAŞ and TEİAŞ companies guided us in terms of legislation and pricing. After our research, we've came up with the result that the changes in both PTF and Brent Oil prices have a significant correlation in the changes of AKSEN sharesi, although the Brent Oil changes apply with a significant delay compared to PTF.

Key Words: Energy, Market Clearing Price, Merit Order, BIST

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LIST OF ABBREVIATIONS

BIST: Borsa İstanbul

PTF: Energy Market Clearance Price, or Piyasa Takas Fiyatı

AKSEN: Aksa Enerji, Aksa Enerji shares in context

EPİAŞ: Energy Exchange İstanbul

1. OVERVIEW

The main goal of this capstone project is to explore the determination of Electricity Market Clearing Price (EMCP, or PTF in Turkish) in the Turkish markets and whether fluctuations in EMCP directly affect BIST energy stock performances. As an exemplary case, the company Aksa Enerji Üretim A.Ş. (AKSEN) was presented. An index for the electricity companies similar to Aksa Enerji Üretim A.Ş. exists in BIST under the name of BIST Electricity (XELKT). Notable companies with similar trade volume as AKSEN in the market are ODAŞ Elektrik (ODAS), Zorlu Enerji (ZOREN) and Enerjisa (ENJSA).

One example of analysis we will conduct in our project is called the merit order module. Broadly, merit order module calculates energy prices based on the demand for power and marginal costs of the electric producers. The merit-order effect, which seeks to increase the penetration of renewable energy, is thought to lower electricity costs. To have a broader approach, we will also conduct a side-by-side comparison with foreign comparable companies to further investigate the global impact of EMCP in energy company shares.

The team consists of three management engineers, Burak Can Ağgül, Melike Koygun and Utku Süleymanoğlu and two energy systems engineers, İlker Eser and Barış Egemen Tokul. Management engineers are expected to examine the relationship between the change in PTF and the financial performance of the firm and stock and energy systems engineering students are to investigate the determinants of Electricity Market Clearing Price (PTF) in Turkish Markets.

1.1. Identification of the need

Purpose of this capstone project is to provide an insight and investment strategy for potential investors of the energy sector. If our analyses provide statistically credible results regarding EMCP prices and company share prices, energy sector investors could have a chance to forecast a trend and build an investment strategy.

1.2. Definition of the problem

To explore the correlation between EMCP and BIST Energy stock prices.

1.2.1. Constraints

Scheduling: 3 management engineering students of the group are expected to play to their strengths of financial analysis and 2 energy system engineering students are expected to handle the examination of the energy market analysis. Time is also of essence in the capstone project, as the

deadlines are tight and group members have internships that take most of their time.

Social Impact: Results of our study might influence investors to invest in energy stocks in contrast to the changes in EMPC. More people being aware of this case might cause fluctuations in the market regardless of the results.

1.3. Literature Review

The energy market settlement handles a variety of concerns with the day-ahead market, intraday market, power socket connections, YEKDEM, electricians, and organized YEK-G owners, as well as any consequent debts or receivables from energy imbalances. The EPA provides a quick, dependable, and transparent answer. The wholesale power service should not profit or lose from these trades. Each lodging trade is customized and reflected in advance allocation as invoice advances, with the possibility to extend the outcome of any commercial usage. The findings of the reconciliation are included in the monthly invoice.

Intraday market efforts to make the Turkish Electricity Market more transparent and effective, financially strong, and integrated with the European Union's electricity markets are ongoing. The Intraday Market, in addition to the existing Day Ahead, Ancillary Services, and Balancing Power Markets, will provide near real-time trading opportunities and allow market participants to balance their portfolios in the short term. According to Yunus (2016) efforts are undertaken in the intraday market to limit the costs of imbalances that may emerge due to unanticipated occurrences (breakdowns, imbalances in renewable power centers, etc.) that may occur in real time.[1]

The purpose of the Day Ahead Market EPA electricity market operation license is to introduce a range of improvements to the way electricity markets are planned, established, developed, and operated, intending to make them more efficient, transparent, and reliable. Additionally, research has shown that the day-ahead market clearing price tends to fluctuate more stable than the balancing power market (Yunus, 2016). [2] These efforts aim to better meet the needs of the electricity sector. The Day Ahead Market and Intraday Market aim to balance production and consumption for the System Operator (MYTM), but real-time deviations can occur. If a power plant has a malfunction or a large consumption facility suddenly starts or stops operation, the balance is disrupted. The NLDC will try to restore balance by using offers from the Balancing Power Market. The DGP generation company uses TPYS (TEİAŞ Market Management System) to inform the System Operator about load-taking and load-shedding offers in the Balancing Power Market. The System Operator arranges these offers by price on an hourly basis. Additionally, the main goal of this market is to quickly balance the system and maintain sufficient supply quality, rather than facilitating trade (Mustafa

Ensari, 2019).[3]

The impact of MCP on energy companies' balance sheets and company values depends on many parameters. One of the important points is which type of energy companies sell to the market. Likewise, the daily total production amount of the company directly affects the balance sheet and this level of influence is directly related to the MCP.

In order to create the MCP price, a market called the Day Ahead Market (DAM) was created in 2011, which is traded one day before the physical electricity delivery. According to Haliloglu and Tutu (2018), the successful forecasting of the energy that will be needed in this market, where market participants share their electricity generation and consumption plans for the next day, prevents the emergence of bottlenecks or energy surpluses.[4] According to EPİAŞ, in the Day-Ahead Market, the buying and selling quantities and market clearing prices are announced in line with the offers submitted by the participants, while the continuous trading method is used in the Intraday Market and the Futures Electricity Market. [5] According to Ata (2016), these prices can be affected by many other predictable external factors, especially the supply-demand balance. According to Tür (2019), the main factors affecting prices can be listed as follows. [6]

- Supply-demand balance and seasonality (non-weekdays and months)
- Installed and generated power
- Power plant type
- Power plant operation policies/input costs/breakdown-maintenance-repair
- Climatic conditions

A typical weekly energy load match amount and market clearing price change in the day ahead market is shown in (the slide) Figure 1. As can be seen here, the fluctuation in price is related to the amount of load.

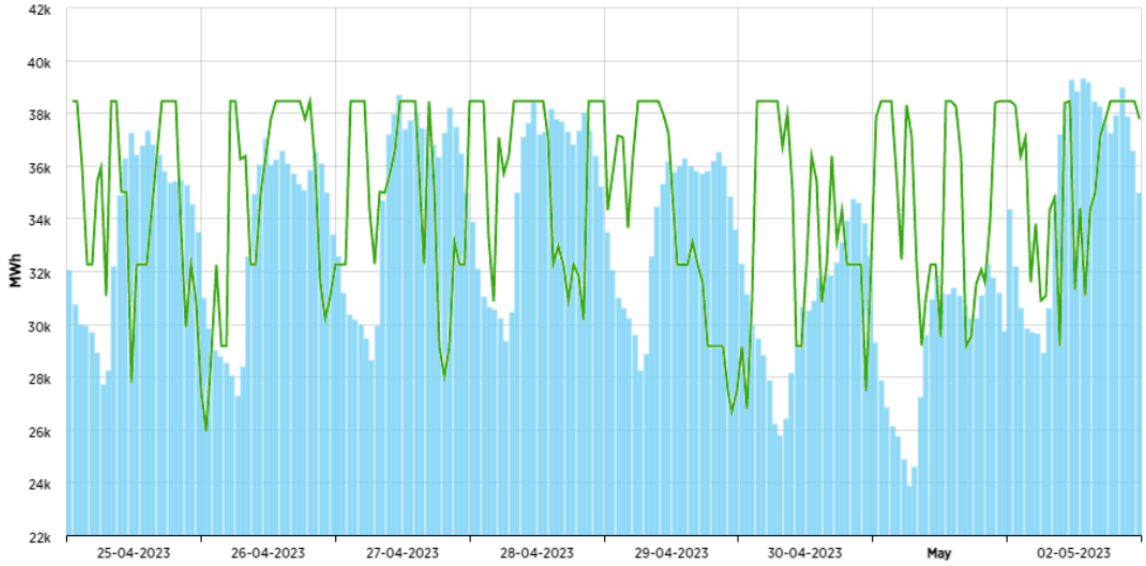

Figure 1. One-week (25 April - 3 May 2023) energy load matching and market clearing price changes (EPİAŞ)*

According to Karagöl and Kavaz (2017), In Turkey, the resources used to produce power are mostly coal and natural gas. Dams and hydroelectric power plants are used to generate electricity after this. The percentage of wind, solar, geothermal, and other energy sources used to generate power is less than 10% [7]. Turkey had produce 34.6% of its power from coal in 2022, 22.2% from natural gas, 20.6% from hydraulic energy, 10.8% from wind, 4.7% from solar, 3.3% from geothermal energy, and 3.7% from other sources (Minister of Energy and Natural Resources, 2022).

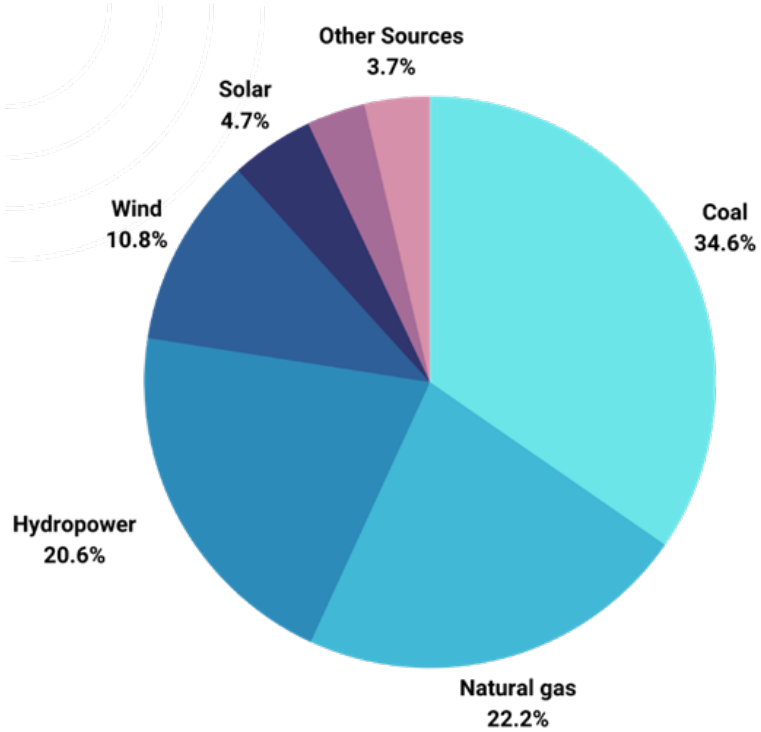

Figure 2. The shares of resources in electricity generation in 2022 (ETKB)*

Although renewable energy technologies don't need fuel and have usually minimal operation and maintenance (O&M) costs, their upfront capital costs are rather expensive for each installed unit of capacity. One of the main obstacles to using renewable technology more extensively is thought to be their high capital costs (Nalan et al., 2009).[8]

According to Nalan et al. (2009) however, the initial capital investment required for renewable technologies is substantial and must be recouped gradually over a long period of time, making it challenging for renewable energy to attract financing. Investment in renewable energy technology is thus first discouraged.[9]

With the development of technology, the rate of electricity consumption is increasing in our country as well as in the whole world. In 2021, Turkey's electrical energy production amounted to 334,723 million kWh with an increase of 28,020,0 million kWh, corresponding to 9.1%, compared to the previous year, and the consumption was 332,871 million kWh, with an increase of 26,762.2 million kWh, corresponding to 8,7%. [10] With the increase in this rate, the competition in the energy sector is increasing in order to meet the required consumption and companies are trying to stand out in this competition with their existing installed power. At the end of 2021, Turkish Electricity Installed Power increased by 4.1% compared to the previous year and reached 99,819.6 MW.[11] The higher the installed power, the more income for the companies, and this reflects positively on their income.

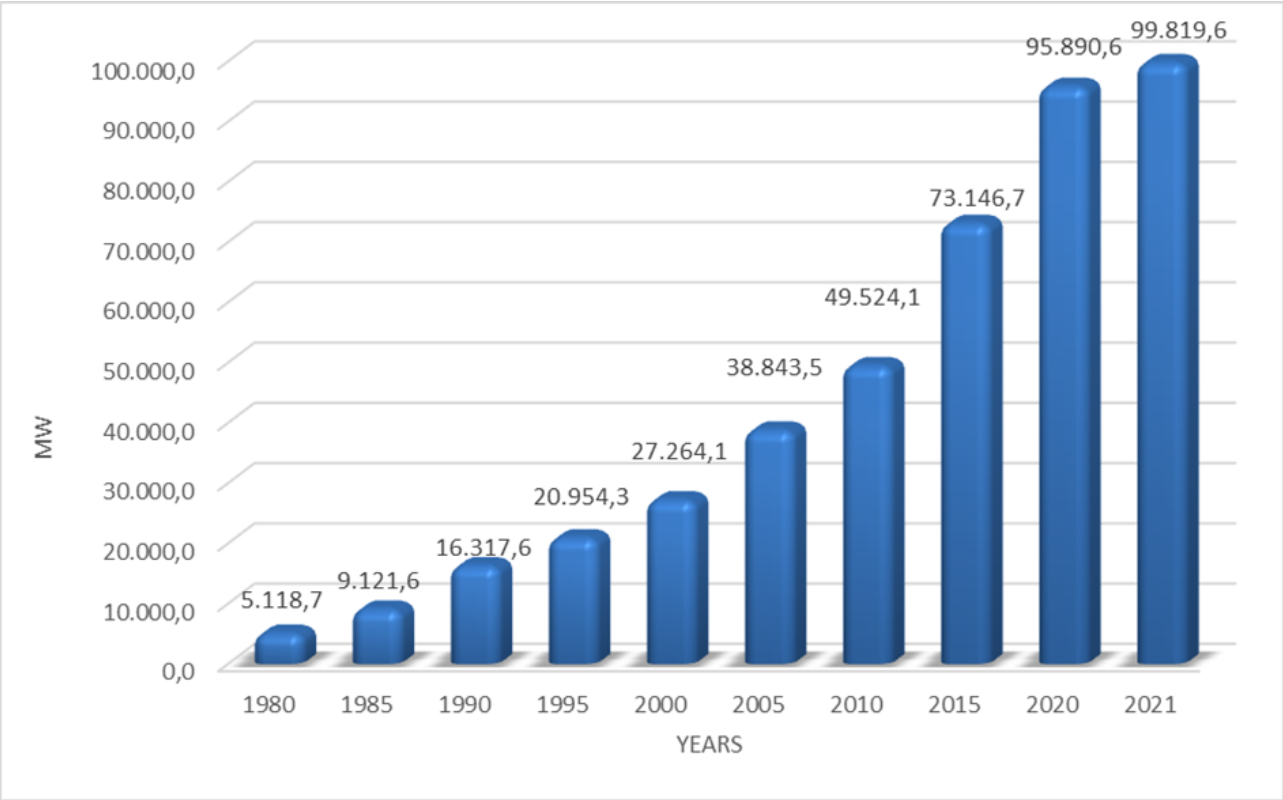

Figure 3. Development of Turkey's installed power over the years. (TEİAŞ)*

The profit calculation of the companies over the MCP price is made as follows; if the PTF is 100 (TL/MWh) for any time zone, producer X has given its offer of 300 MWh from 50 (TL/MWh) days in advance for that hour and the given offer has been evaluated. In this case, producer X gets (100-50)*300=15000 TL profit.

According to Berk and Torun (2019), the marginal generation cost and consequently the prices in the energy system are projected to fall as the share of renewables increases in contrast to fossil fuels, so energy economics literature refers to this as the merit-order effect. [12] According to Berk and Torun (2019), in order to determine whether the merit-order effect—that is, the consequences of growing wind penetration on day-ahead market clearing prices—exists in the Turkish energy market, this article will test its existence. [13] According to Berk and Torun (2019), the electrical market in Turkey has changed significantly in recent years. The 2001 Electricity Market Law (Law No. 4628) not only established a firm foundation for a more competitive market but also Established government support measures for the Turkish electricity sector's use of renewable energy potential, because the marginal cost of producing electricity from renewable sources is so much less than the cost of producing electricity from thermal power plants that burn fossil fuels, particularly coal and natural gas, increasing the production of electricity based on renewable sources would result in lower electricity prices. [14]

Since there isn't enough study made on the electricity market in contrast to the electricity stock index, a bulk of the literature we reviewed had to be about other commodities, especially oil. We then came to the conclusion that there is, even in the slightest, an existing relationship between oil prices and the stock market, which then led us to believe there should be a relationship between electricity market and stock market as well. Comparing other commodities with both Turkish and global indexes, Faff and Brailsford (1999) discovered a positive interaction between oil and gas industry stock values and monthly oil prices in their analysis of the Australian industry sector and monthly oil prices. [15] Akgün (2006) found that there is a positive relationship between the IMKB-100 (old BIST) index and oil prices as a result of his study with the IMKB-100 index and oil price data for the years 1997-2004. [16] Güler, Ramazan, and Orçun (2010) discovered that variations in Brent oil prices from 2000 to 2009 have had an impact on stock prices in the said time frame. [17] Kapusuzoğlu (2011) used Granger causality tests and Johansen cointegration analysis to evaluate the link between Brent oil prices and the IMKB 100, IMKB 50, and IMKB 30 indexes for the years 2000–2010. The author concludes the analysis by asserting that all indicators and oil prices have a long-term association. The Granger causality test revealed a one-way causal relationship linking all indices to oil prices. [18]

Below is a bundle of literature which was included in the report written by Akhbulaev et al. (2022) [19], which shows considerable amount of literature points to an existing relationship, with a few exceptions:

Berk and Aydoğan (2012)	Turkey	Oil	1990-2011	Relationship exists
Ünlü and Topçu (2012)	Turkey	Oil	1990:1-2001:2 ve 2001:3-2011:12	Relationship exists
Eryiğit (2012)	Turkey	Oil	2005-2008	Relationship exists
Acaravcı, Öztürk and Kandır (2012)	15 European countries	Natural gas	1990:1-2008:1	Relationship exists
Ratti and Hasan (2013)	Australia	Oil	31 Mart 2000-31 Aralık 2010	Relationship exists
Antonakakis ve Filis (2013)	USA, UK, Germany, Canada, Norway	oil	1988-2011	no relationship
Chortareas and Noikokyris (2014)	USA	Oil	1981-2006	Relationship exists
Zortuk and Bayrak (2015)	G-7	Oil	2002-2014	Relationship exists
Bolaman Avcı (2015)	Turkey	Oil	2003-2013	Relationship exists
Yalçın (2015)	Russia, Kazakhstan and Ukraine	Oil	2000-2013	Relationship exists
Yıldırım (2016)		Oil	2003:1-2016:1	Relationship exists
Bastianin, Conti and Manera (2016)	G-7	Oil	1973-2015	Relationship exists
Sandal et al. (2017)	Turkey	Oil	2005:01-2015:12	no relationship
Karhan and Aydın (2018)	Turkey	Oil	05.01.2009-6.08.2018	no relationship
Kumar et al. (2019)				Relationship exists
Dursun and Özcan (2019)			2005-2017	Relationship exists
Akbulaev and Rahimli (2020)	India	Oil and natural gas	2007-2019	Relationship exists
Suleymanli et al. (2020).		Oil and natural gas	2008-2019	Relationship exists
Sakinç (2021)	25 OECD member countries	Oil	21 Ağustos 2020	Relationship exists
Karakuş (2021)	Russia	Oil and natural gas	2010-2019	Relationship exists

Figure 4. Relationship between different commodities and different stock markets

Some of the models that were used in these sources are explained in the third section.

Economic activity and climate variables both influence electricity demand. However, the demand is inelastic. When demand is low, manufacturers use their existing capacity to create power at low marginal costs. In the winter and summer, when demand increases on specific days of the week or even at certain times of the day, manufacturing units with high marginal costs are activated. Lucia and Schwartz (2002) discovered seasonal influences on spot and future contract prices in the Nord Pool, and this is a simple sinusoidal function. [20] The weekly and monthly impacts were specifically explored in this study. With an increase in demand, power facilities that provide high-cost output will come online, causing prices to rise. Prices will begin to revert to pre-decommissioning levels once such power plants or manufacturing units are decommissioned. The propensity to revert to the mean explains this predicament.Research has explained the degree of tendency to return to the average in the formation of prices with the characteristics of time-dependent conditional volatility and sudden price increases. The volatility of electricity prices varies with time in relation to fluctuations in supply and demand and climatic conditions.

According to Joseph Cullen (2013), marginal cost of electricity generation, which is determined by the cost of fuel, operation and maintenance charges, and the efficiency of the power plants, is one of the important elements influencing the PTF.[21] As the marginal cost of producing electricity fluctuates, so does the market clearing price, which impacts the profitability and market value of energy firms. According to Hirth (2013), when fuel prices are high, the marginal cost of electricity generation rises, causing the PTF to rise. This can boost the market value of energy companies with low-cost generation assets by increasing their profit margins. According to Hirth (2013), energy businesses with high-cost generation assets, on the other hand, may see their market value fall if their profit margins contract due to higher costs.[22] According to Mutlu Yılmaz (2012), precipitation is very important for energy production from hydraulic power plants in semi-arid countries such as Turkey, and the occupancy rate of the dams and the amount of energy produced decrease in dry years.[23]

The Turkey Electricity Market Settlement (TEIAS) is a government-run entity responsible for managing the electricity market in Turkey. TEIAS is the main clearinghouse for electricity transactions in the country, ensuring that all parties involved in the market settle their financial obligations accurately and on time. TEIAS works by calculating and distributing payments between electricity generators, distributors, and retailers in Turkey. According to Ercüment Camadan &

Ibrahim Etem Erten (2011) the process begins with the electricity generators, who produce electricity and sell it to the wholesale market.[24] The wholesale market then sells the electricity to distribution companies, who transport the electricity to end-users. Finally, the retailers sell the electricity to end-users and bill them for their consumption. TEIAS serves as an intermediary in this process, ensuring that all parties involved in the market settle their financial obligations accurately and on time. According to Ercüment Camadan & Ibrahim Etem Erten (2011) does this by monitoring the amount of electricity generated, sold, and consumed, and calculating the financial obligations of each party based on their contracted agreements.[25] TEIAS also ensures that the market operates efficiently by monitoring and regulating the supply and demand of electricity. For example, if the demand for electricity is high, TEIAS may offer incentives for generators to produce more electricity to meet the demand. Similarly, if there is excess supply, TEIAS may offer incentives for distributors to increase their sales. Overall, the Turkey Electricity Market Settlement plays a critical role in ensuring that the electricity market in Turkey operates smoothly and efficiently, and that all parties involved in the market settle their financial obligations accurately and on time.

Relationship Between Energy Market Settlement and Market Clearing Price

In the energy market, settlement refers to the process of calculating and reconciling payments between buyers and sellers for the energy that has been traded. The settlement process typically occurs after the energy market has closed for the day, and it is an important step in ensuring that all participants in the market receive the correct payments for their energy transactions. The market clearing price (MCP) is the price at which all energy offered for sale in the market is purchased. Kwame Richard Klu & Kwame Yaro Appiah (2020) ıt represents the equilibrium point at which the total quantity of energy demanded by buyers is equal to the total quantity of energy supplied by sellers.[26] The MCP is determined by the intersection of the supply and demand curves in the energy market. The settlement process and the MCP are closely related because the settlement price is often based on the MCP. For example, in a market where energy is traded at a fixed price, the settlement price will be the same as the MCP. However, in a market where energy is traded at different prices depending on the time of day, location, or other factors, the settlement price may be different from the MCP. In such cases, the settlement price is calculated by using a formula that considers the energy that has been traded at different prices and the corresponding quantities. The settlement price is typically weighted based on the volume of energy that has been traded at each price point, and it is often calculated using complex algorithms and software systems. Overall, the

settlement process and the MCP are both important factors in the energy market because they help ensure that buyers and sellers receive fair compensation for their energy transactions, and they help maintain a stable and efficient energy market. By accurately calculating the settlement price based on the MCP, market participants can have confidence in the reliability and transparency of the energy market, which can encourage further investment and growth in the industry.

The profit calculation of the companies over the MCP price is made as follows; if the PTF is 100 (TL/MWh) for any time zone, producer X has given its offer of 300 MWh from 50 (TL/MWh) days in advance for that hour and the given offer has been evaluated. In this case, producer X gets (100-50)*300=15000 TL profit.

2. WORK PLAN

Our project, named relationship between electricity market clearing price (PTF) and market value of energy companies, will be done by management engineering students Burak Can Ağgül, Melike Koygun and Utku Süleymanoğlu, and energy systems engineers İlker Eser and Barış Egemen Tokul. Timeframe of our analysis spans from February to June.

2.1 Work Breakdown Structure (WBS)

The project will be separated into two sub-parts for management engineers and energy systems engineers to have a fair planning regarding their skills. First subpart will be done by management engineers and the second subpart will be carried out by the energy systems engineers.

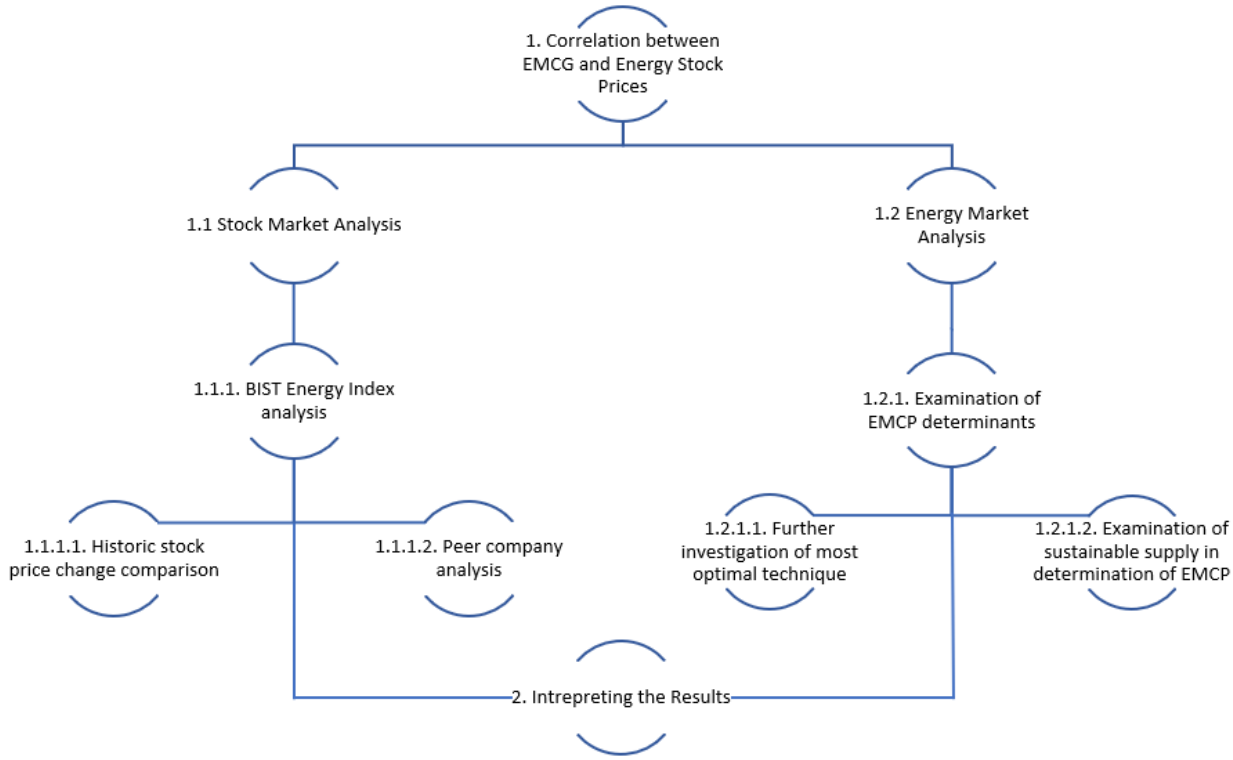

Figure 5. Work Breakdown Structure

2.2 Responsibility Matrix (RM)

First subpart of the project, or the financial analysis part, will be carried out by management engineers Burak Can Ağgül, Melike Koygun and Utku Süleymanoğlu, and energy market analysis part will be carried out by energy systems engineers Barış Egemen Tokul and İlker Esen. Main responsible for planning of report structure, meetings, workload management and task tracking is Utku Süleymanoğlu, with Burak Can Ağgül and Melike Koygun as support. Joint responsibles of integration of financial analysis and energy market analysis are Burak Can Ağgül and Barış Egemen Tokul, with all members in support. Joint leads in documentation of the report are Melike Koygun and İlker Eser, with all members in support.

Task	Burak	Melike	Utku	Barış	İlker
Financial Analysis	**R**	**R**	**R**		
Energy Market Analysis				**R**	**R**

Planning	S	S	**R**		
Integration	**R**	S	S	**R**	S
Documentation	S	**R**	S	S	**R**

R: Responsible S: Support

Table 1. Responsibility Matrix

2.3 Project Network (PN)

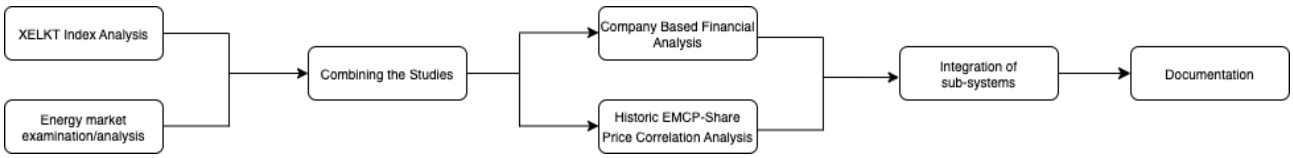

Figure 6. Project Network

2.4 Gantt Chart

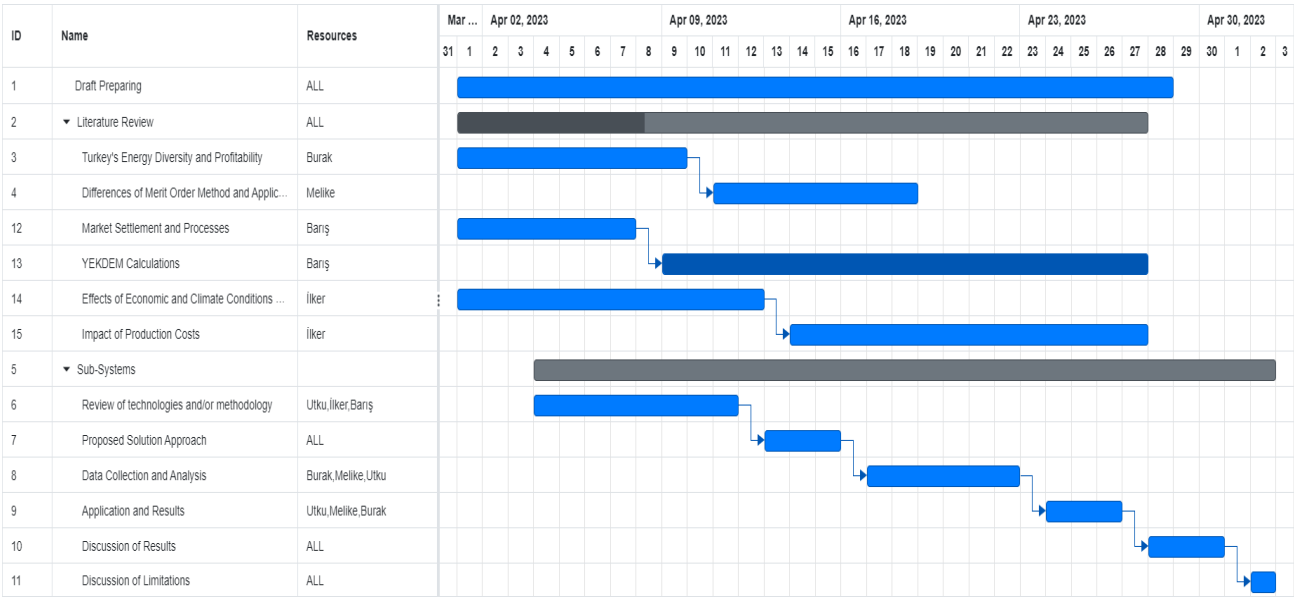

Figure 7. Gantt Chart

2.5 Risk Assessment

RISK LEVEL		Severity of the event on the project success		
		Minor	Moderate	Major
Probability of the event occuring	Unlikely	VERY LOW	LOW	MEDIUM
	Possible	LOW	MEDIUM	HIGH
	Likely	MEDIUM	HIGH	VERY HIGH

VERY LOW	This event is very low risk and so does not require any plan for mitigation. In the unlikely event that it does occur there will be only a minor effect on the project.
LOW	This event is low-risk; a preliminary study on a plan of action to recover from the event can be performed and noted.
MEDIUM	This event presents a signficant risk; a plan of action to recover from it should be made and resources sourced in advance.
HIGH	This event presents a very signficant risk. Consider changing the product design/project plan to reduce the risk; else a plan of action for recovery should be made and resources sourced in advance.
VERY HIGH	**This is an unacceptable risk. The product design/project plan must be changed to reduce the risk to an acceptable level.**

Table 2. Risk Matrix

Failure Event	Probability	Severity	Risk level	Plan of Action
Spot Electricity Market Risk	**Possible** It happens as a result of sudden changes in supply or demand.	**Major** It will produce significant market volatility and disruptions.	**HIGH**	The market is controlled monitored, and the source of the volatility is identified. Following the identification of the causal cause, preventative steps are implemented.

YEKDEM Risk	**Possible** Many variables influence the unit cost of YEKDEM, including legislative changes, spot market rates, power plant capacity factors, and the dollar exchange rate. As a result, changing the price of YEKDEM entails risks.	**Major** It creates large fluctuations in the currency rate. Furthermore, it leads to high-priced bills for both the manufacturer and the user.	**HIGH**	YEKDEM risk originating from spot prices can be hedged over the portfolio's closing rate.
Increasing adoption of renewable energy sources	**Possible** Renewable energy sources such as wind and solar power are becoming increasingly cost-competitive with traditional fossil fuels	**Major** a decrease in the PTF and a corresponding decrease in the revenues and profits of energy companies that rely heavily on fossil fuels	**HIGH**	Diversify energy portfolio to include more renewable sources, invest in R&D to improve renewable energy technologies, and lobby for favorable regulatory policies that support renewable energy development.
Changes in regulatory policies	**Possible** Regulatory policies can change as a result of political and economic factors	**Major** Changes in regulatory policies can affect the PTF and the market value of energy companies by affecting the cost of electricity production, revenues, and profits	**MEDIUM**	Monitor regulatory developments, lobby for favorable regulatory policies, and diversify energy portfolio to include sources that are less affected by regulatory changes.

Competition from other companies	**Possible** Competition from other companies is a constant risk in any industry	**Major** Increased competition can lead to a decrease in the PTF and a corresponding decrease in the revenues and profits of energy companies	**MEDIUM**	Monitor the competition, invest in marketing and sales to increase brand awareness and customer loyalty, and invest in R&D to develop new and innovative products and services that differentiate the company from its competitors.
Cybersecurity threats	**Possible** Cybersecurity threats are becoming increasingly common in all industries	**Major** Cybersecurity breaches can lead to the loss of sensitive data and assets, and can disrupt electricity production and distribution	**MEDIUM**	Invest in cybersecurity measures such as firewalls, intrusion detection systems, and employee training, and implement incident response plans to minimize the impact of any potential breaches.

Table 3. Risk Assessment Table

- **3. Management Engineering - Energy Systems Engineering**

For the 3rd section, both departments are included in the collaborative description of the work that is required to be completed for this research.

3.1. Review of technologies and/or methodology

To conduct most of our analysis, for both financial and electric market parts, Excel and R-Studio used extensively throughout our project. Also, several calculation models are covered in the project.

Excel: In this project Excel was used to analyze the collected data.

R Studio: R Studio was used for understanding the collected data and finding a result.

3.2. Proposed solution approach

The Borsa Istanbul, or BIST, is the singular entity of exchange in Turkey. Under it, there are a myriad of stock indexes, with XELKT index, or BIST Electricity Index being one of them. The standing value of a stock exchange index is calculated by this formula:

$$Et = \Sigma (Fit \times Hit \times Nit) / Bt$$

Where:

Et = Index Value at time "t"

Fit = Price of share "i" at time "t"

Hit = Volume of share "i" at time "t"

Nit = Total amount of share "i" at time "t"

Bt = Base value of index at time "t"

Under the XELKT index umbrella, by May 2023, there are 26 companies. These being, historically sorted:

AKSEN	Aksa Enerji	**AYDEM**	Aydem Enerji
ZOREN	Zorlu Enerji	**CANTE**	Can2 Termik
ODAS O	Odaş Enerji	**KARYE**	Kartal Enerji
AYEN	Ayen Enerji	**MAGEN**	Margun Enerji
AKENR	Ak Enerji	**ARASE**	Doğu Aras Enerji
AKSUE	Aksu Enerji	**HUNER**	Hun Enerji
ENJSA	Enerjisa	**SMRTG**	Smart Güneş Enerjisi
NATEN	Naturel Enerji	**CONSE**	Consus Enerji
ESEN	Esenboğa Elektrik	**ZEDUR**	Zedur Enerji
PAMEL	Pamel Elektrik	**ALFAS**	Alfa Solar Enerji

NTGAZ	Naturelgaz	**AHGAZ**	Ahlatcı Doğalgaz
GWIND	Galatawind	**AKFYE**	Akfen Yenilenebilir Enerji
BIOEN	Biotrend	**CWENE**	CW Enerji

With CW Enerji added to the roster last, on May 4th.

We based our research on Aksa Enerji, (BIST Code: AKSEN). Established in 1997, Aksa Energy is Turkey's largest free-standing electricity producer. The primary partner of Aksa Enerji, Kazancı Holding, is a multinational corporation with operations in 22 countries, production on four continents, and sales to 178 nations. It was established in the 1950s. Through its subsidiaries, Kazancı Holding, one of the top organizations in the energy sector, produces and sells energy, distributes and sells electricity and natural gas, and produces and sells generators, each of which is a leader in its industry. The Group generates value through its interests in agriculture and tourism in addition to energy. In 2010, Aksa Enerji shares began to be traded on Borsa Istanbul with the code AKSEN. Aksa Energy shares; BIST 30 is traded in BIST Electricity, BIST Corporate Governance Index and MSCI Turkey Index.

Figure 8. Logo of Aksa Enerji.

In Turkey's energy market, the day-ahead market is the primary market where buyers and sellers trade electricity for delivery on the next day. The day-ahead market determines the market clearing price (MCP) which is the price at which the demand for electricity equals the supply. The MCP is calculated as the price of the highest accepted bid, which clears the market. The intraday market, on the other hand, is a secondary market where market participants can buy and sell electricity on the day of delivery, after the day-ahead market has closed. The intraday market allows market participants to adjust their positions and respond to changes in demand and supply, which may not

have been accounted for in the day-ahead market. The introduction of the intraday market in Turkey has increased the liquidity of the market and provided more opportunities for market participants to trade electricity. İbrahim Murat BİCİL (2015) intraday market has helped to reduce the volatility of the day-ahead market prices by allowing market participants to hedge their positions and balance their portfolios.[28] The effect of intraday and day-ahead markets on the MCP in the Turkish energy market depends on the level of demand and supply, as well as the availability of generation capacity. When demand is high and supply is tight, the MCP is likely to be higher in both the day-ahead and intraday markets. Conversely, when demand is low and supply is plentiful, the MCP is likely to be lower in both markets. Overall, the intraday market provides an additional layer of flexibility and responsiveness to the Turkish energy market, allowing for more efficient price discovery and market clearing.

In the Turkish energy markets, there is a close relationship between the Renewable Energy Resources Support Mechanism (YEKDEM) and the market clearing price. YEKDEM is a support mechanism where the electricity produced from renewable energy sources is purchased at a price above the market clearing price. This mechanism aims to promote the expansion and use of renewable energy sources. The implementation of YEKDEM is directly related to the market clearing price. Since the price to be paid for YEKDEM is higher than the market clearing price, an increase may be seen in the market clearing price when this support mechanism is used. The reason for this is that as a result of the widespread use and use of renewable energy sources, the supply-demand balance in the market will be affected. More supply of electricity produced from renewable energy sources to the market may cause prices to fall, while less supply to the market may cause prices to rise. For example, if Turkey's electricity needs are met by electricity generated from renewable energy sources such as solar or wind energy for one day, the market clearing price may decrease. However, when electricity demand is high, the market clearing price may increase. In addition to the increase in the market clearing price with the support of YEKDEM, the increase in the use of renewable energy sources provides an environmentally friendly approach to meeting energy needs. For this reason, there is a direct relationship between YEKDEM and the market clearing price, and this relationship becomes more important with the widespread use and use of renewable energy sources.

Total Debt Item for Yekdem

$$YEKTOB = \left(\sum_{i=1}^{a}\sum_{b=1}^{n}\sum_{t=1}^{l}\sum_{u=1}^{k}\left(UEVM_{i,b,t,u} \times y_{i,b,t,u} \times YEKF_{i,b} \times KUR_u\right)\right) + \sum_{j=1}^{m} LÜYTOB_j$$

Figure 9. Total Debt Item for YEKDEM

Total Credit Item for Yekdem

$$YG = \sum_{i=1}^{a}\sum_{b=1}^{n_i}\sum_{t=1}^{l}\sum_{u=1}^{k}\left(UEVM_{i,b,t,u} \times y_{i,b,t,u} \times PTF_{t,u} \times j\right) + \sum_{m=1}^{d}\sum_{t=1}^{l}\sum_{u=1}^{k}\left(LÜNET_{m,t,u} \times PTF_{t,u} \times p\right) + \sum_{c=1}^{g} KB_c + \sum_{m=1}^{h} FB_m$$

Figure 10. Total Credit Item for Yekdem

Coefficient to be Determined for Capacity Increase

$$kg_{i,b,u} < ig_{i,b,u} \quad \text{if} \quad y_{i,b,t,u} = \frac{kg_{i,b,t,u}}{ig_{i,b,t,u}}$$

$$kg_{i,b,u} \geq ig_{i,b,u} \quad \text{if} \quad y_{i,b,t,u} = 1$$

Figure 11. Coefficient to be Determined for Capacity Increase

Pay Liability Ratio for Yekdem

$$ÖYO_i = \frac{\sum_{b=1}^{n}\sum_{u=1}^{k} UEÇM_{i,b,u}}{\sum_{b=1}^{m}\sum_{u=1}^{k} UEÇM_{b,u}}$$

$$ÖYT_i = YEKTOB \; x \; ÖYO_i$$

$$YGT_i = YG \; x \; ÖYO_i$$

Figure 12. Pay Liability Ratio for Yekdem

Arbitrage Model:

Arbitrage refers to the strategy of earning risk-free profit by capitalizing on price discrepancies. The arbitrage model assumes that the market exchange rate should reflect these disparities in different

markets or asset classes. There are some fundamental formulas for the arbitrage model, which include: Price Discrepancy: It represents the difference in prices in different markets or asset classes. Arbitrage opportunities arise from these price discrepancies. The formula for price discrepancy can be expressed as follows:

Price Discrepancy = Price1 - Price2

Arbitrage Spread: This measures the potential profit resulting from an arbitrage opportunity. The arbitrage spread is calculated by considering the costs related to the price discrepancy and transaction expenses.[29] The formula for arbitrage spread is as follows: Arbitrage Spread = Price Discrepancy - Transaction Costs – Costs The arbitrage model aims to identify risk-free arbitrage opportunities by analyzing price discrepancies and the arbitrage spread."

The Markov chain model is a statistical tool for forecasting future price changes. In this concept, prices are represented as a chain that can alternate between various circumstances over a predetermined amount of time.Dec. The Markov chain model has the following fundamental formula:

$P(X_{t+1} = x \mid X_t=x_t, X_{t-1} = x_{t-1}, ..., X_1=x_1) =P(X_{t+1}=x|X_t=x_t)$

X_t represents the current price state, X_t represents the future price state, and x represents the possible values for the price state in this formula. The likelihood of changing from one price condition to another is determined using this formula. First, transition probabilities from previous price movements are computed for the Markov chain model. Then, by understanding the present price environment, future price fluctuations are projected. By starting from the current condition and working our way through the chain of events, these forecasts are made.

Marshall's Price Elasticity in Percentage: The supply and demand functions are used in the model. Price is frequently used to express supply and demand functions. The swap price is established in this model by locating the point where the supply and demand curves overlap.[30]

Marshall's Percent Price Elasticity Model has the following mathematical formula: Price Elasticity = $(\Delta Q/Q) / (\Delta P/P)$

Here:

ΔQ → represents the change in quantity demanded.

Q → represents the initial quantity demanded.

ΔP → represents the change in price.

P → represents the initial price.

The change in quantity required is given as a percentage of the change in price in the price elasticity formula. This ratio represents the price elasticity. Supply and demand curves are used to compute the swap price in Marshall's percent price elasticity model. The price elasticity coefficient is derived at the intersection of the curves, and how supply and demand quantities will react to price changes is studied using this coefficient. Assume that as the quantity required for an item rises by 10%, the price rises by 5%. In this situation, Marshall's percent price elasticity calculation says:

Price Elasticity = (0.1/1) / (0.05/1) = 2

If the estimated value of E equals one, it denotes unitary price elasticity of demand, in which the percentage change in quantity requested is proportionate to the percentage change in price.

Assume the initial quantity required (Q) is 100 units and the initial price (P) is $10. The new price (P') after a price adjustment is $12, and the new quantity requested (Q') is 80 units. Inputting the following values into the formula:

E = [(80 - 100)/100] / [(12 - 10)/10]

E = (-20/100) / (2/10)

E = -0.2 / 0.2

E = -1

The determined value of E in this situation is -1, suggesting that demand for the product is price elastic, since a 1% rise in price results in a 1% drop in quantity desired. It's vital to remember that Marshall's percent price elasticity model assumes ceteris paribus, which means it takes into account demand price elasticity while maintaining all other variables constant. In actuality, elasticity of demand for a commodity may be influenced by a variety of factors such as income, alternatives, complements, and customer preferences.

Count Model:

The Cournot Model is a mathematical model used to predict the output of enterprises that compete in oligopolistic markets. While each business strives to maximize its profit in this model, it also

considers the outputs of other enterprises. The mathematical formula and explanation for the Cournot Model are as follows:

Assumptions:

The oligopoly market has N businesses.

Each business takes the production volumes of the other firms as fixed.

Each corporation selects its own production in order to maximize profits.

Production Function: A production function is used by each organization to calculate the amount of production. Typically, a linear production function is utilized. For instance, i. The firm's production function may be stated as:

$Qi = a - b * (Q1 + Q2 + ... + Qi-1 + Qi+1 + ... + QN)$

Here:

Qi →represents the firm's output.

a → denotes the maximum production quantity.

b → is a parameter that shows the sensitivity to the production amount of other firms.

Profit Function: Each business decides how much to produce in order to maximize its earnings. Production expenses and pricing are included in the profit function. For instance, i. The profit function of a company may be stated as follows:

$\pi i = P * Qi - C(Qi)$

Here:

πi → represents the firm's profit.

P → stands for the price in the market.

C(Qi) → It represents the cost function that depends on the firm's output.

Profit Maximization: Every company optimizes its profit function in order to maximize profits. The volume of output that will maximize earnings is determined by the company. The first derivative of the profit function is utilized for this. Using the first derivative, the profit-maximizing output is:

$$\partial \pi i/\partial Qi = P - \partial C(Qi)/\partial Qi - b * P$$

This is the fundamental equation that permits enterprises to calculate their own production numbers. Each corporation determines the amount of production that will optimize its earnings using this equation. The Cournot Model encourages businesses to deliberately measure their output in order to optimize profitability.

Volume Weighted Average Price Model:

Volume Weighted Average Price (VWAP) is a trading indicator that computes the average price of a financial instrument over a certain time period, weighted by trading volume at each price level. It is often used by traders and investors to calculate the average price at which a certain securities has traded during the day or any other time frame.[31]

To calculate the VWAP, you need the following data:

Price: The price at which each trade occurred.

Volume: The number of shares or contracts traded at each price level.

Here's a step-by-step process to calculate VWAP:

Determine the time span: Choose the time interval during which the VWAP should be calculated. For example, you may compute the VWAP for a single trading day, from market open to market close. Add up the volume and price for each trade: For each trade over the chosen time period, add the volume and multiply it by the associated price.

Example:

Trade 1: Price = $100, Volume = 200

Trade 2: Price = $102, Volume = 150

Trade 3: Price = $98, Volume = 100

Total volume = 200 + 150 + 100 = 450

Total value = (100 * 200) + (102 * 150) + (98 * 100) = 20,000 + 15,300 + 9,800 = 45,100

Calculate the cumulative volume and cumulative value: As you progress through the deals, calculate the cumulative total of the volume and value at each price level.

Cumulative volume:

Trade 1: 200

Trade 2: 200 + 150 = 350

Trade 3: 350 + 100 = 450

Cumulative value:

Trade 1: 20,000

Trade 2: 20,000 + 15,300 = 35,300

Trade 3: 35,300 + 9,800 = 45,100

To calculate the VWAP, divide the cumulative value by the cumulative volume.

VWAP = Cumulative Value / Cumulative Volume

= 45,100 / 450

= $100.22

The VWAP in this case is $100.22. It represents the security's average price throughout the chosen time period, weighted by the trading volume at each price level. Traders frequently compare the current market price to the VWAP to see if the security is trading above or below the average price, which may assist them make trading choices.

Merit Order Methodology: Merit Order Methodology is a method to determine which energy sources are firstly used based on variable costs. Variable costs are fuel cost, maintenance costs and other operational costs.

According to Merit order methodology, energy sources with the lowest variable costs are used first, followed with other energy sources with the lowest variable costs.

Listing energy sources in this way, merit order methodology aims to decrease variable costs and greenhouse gas emissions. Also helps to provide reliable and sustainable energy supply.

This methodology is commonly used in most of the countries all around the world.

For example, in an electricity market, the electrical energy demanded for one day is 1000 MW. Let there be 3 different energy sources in this market: wind energy, hydroelectric power and thermal energy produced from natural gas.

Variable costs for each of these energy sources be as follows:

- Wind energy: 0 TL/MWh (there is no fuel cost in wind energy generation)
- Hydroelectric energy: 100 TL/MWh
- Thermal energy produced from natural gas: 150 TL/MWh

In this case, the sources that produce energy with the lowest variable cost by applying the merit order method are put into use first. Therefore, wind energy will be used primarily because its variable cost is zero. If wind power is not enough to meet demand, hydroelectric power will be used because thermal energy has a higher variable cost.

Energy Source	Variable Cost (TL/MWh)	Production Amount (MW)	Total Cost (TL)
Wind energy	-	500	-
Hydroelectric energy	100	400	40.000
Thermal energy	150	100	15.000
Total		1.000	55.000

Figure 13. Variable costs of energy sources

Since the demand is 1000 MW, first 500 MW of wind energy is used, then 400 MW of hydroelectric energy is used and the remaining 100 MW of demand is met from thermal energy. The total cost will be 55,000 TL. In this way, by applying the merit order method, energy sources with the lowest variable cost are always used first and the energy market is operated more efficiently.

Here is the detailed explanation of the formula:

Calculate the total variable cost for each production resource:
TVC = (Fuel Cost + Variable O&M Cost) * Power Output

Fuel Cost: The cost of fuel consumed by the production resource.
Variable Operating and Maintenance Cost: Variable costs specific to the production resource, maintenance and operating cost.
Power Output: The electrical power produced by the generating source.
Determine the capacity factor (CF) for each production resource:
CF = Actual Power Output / Maximum Power Output

Actual Power Output: Electricity produced by the generating source.
Maximum Power Output: The maximum power capacity of the generation source.
Calculate the merit order effect (MOE) for each production resource using the following formula:
MOE = TVC * CF

Arrange production resources in ascending order according to merit order effect values.Generation resources with a lower cost and higher capacity factor will have a greater merit order effect. When these low cost resources are used primarily, they help lower the overall market price of electricity.

There are several factors that affect the merit order method:

Marginal Costs: The marginal cost of each power production source is the primary determining factor in the rank of merit technique.
Fuel Prices: The merit order can be impacted by changes in fuel prices, such as those for coal, natural gas, or oil.

Variable O&M Costs: Resources with lower variable costs are usually used earlier.

Startup and Shutdown Costs: Resources with lower startup costs or faster commissioning times can be advantageous.

Capacity Factors: Resources with a higher capacity factor are usually used first.
Environmental Factors: Regulations or incentives may require or encourage the use of renewable energy sources.

System Constraints: The distribution order within the merit order may be impacted by transmission capacity constraints within the power system or operating limitations of certain switchboards.

Market Design: Certain market rules and mechanisms, particularly capacity payments or different pricing structures, can affect the merit order method.

In order to explore the correlations in a correct manner, we need to first explore the reasoning behind PTF pricing to relate it to the electricity stock market index. After inspecting the different pricing methods in the Turkish energy market, different methodologies are used to correlate the findings with the fluctuations in the stock market. In similar projects, we can see that there are different models to correlate commodity price changes with their respective stock price changes. Some of the examples are:

Granger causality model: Granger causality model is used to determine whether a series of data is useful in forecasting another. This model assesses the direction of causation between two time series of data and is used frequently to shed light on the relationship between stock prices and commodities prices.

Vector Aggression (VAR) model: The VAR model is a statistical time series model that may be

used to examine the correlation between a number of different variables, including the prices of stocks and commodities. The VAR model provides insight into the dynamics and interdependencies of the commodities and stock markets by evaluating these variables collectively. The VAR model is made out of two equations:

$$x_t = c_1 + \sum_{i=1}^{3} \alpha_{1,i} y_{t-i} + \sum_{i=1}^{3} \beta_{1,i} x_{t-i} + \epsilon_{x,t}$$
$$y_t = c_2 + \sum_{i=1}^{3} \alpha_{2,i} y_{t-i} + \sum_{i=1}^{3} \beta_{2,i} x_{t-i} + \epsilon_{y,t}$$

Figure 14. VAR Equations

Granger causality suggests that if the past values of one variable provide significant information for predicting another variable, then the former variable is said to "Granger-cause" the latter variable. In the context of commodity values and a stock's market value, Granger causality analysis can help determine whether past changes in commodity values provide useful information for predicting future changes in the stock's market value.

3.3. Data collection and analysis

Our data was collected from government entities such as KAP, SPK and EPİAŞ, and trusted outside resources such as investing.com and Bloomberg. EPİAŞ used for the energy side of our analysis, and

other entities helped our project with the company based financial part of the analysis.

3.4. Application and results

We carried out our research conjoined as both departments. As a result of our research we gain data from government entities such as KAP, SPK and EPİAŞ, and trusted outside resources such as investing.com and Bloomberg. This data was analyzed with excel tools by management engineers. Result part of the project announced in the other parts of the project.

For our comparison, we used Granger Causality Model and VAR Model to correlate AKSEN data and PTF data, starting from January 1st, 2019, and May 1st, 2023 on Python. The Granger model, specifically Granger causality, is a statistical concept used to analyze the causal relationship

between two time series variables. It can be applied to examine the relationship between commodity values and a stock's market value.

The first step in using the Granger model is to collect historical data for the market value of the stock and the commodity value over a given time frame. In our case, we collected AKSEN share prices and PTF prices, and collected them daily since the Granger model needs a specific time frame to work.

In order to evaluate the relationship between the two variables, we used a time series analytic method, specifically vector autoregression (VAR) . In order to accurately represent the temporal dynamics, the VAR model uses the lagged values of both variables as predictors. The reason of lagged value usage is the changes in the value of PTF would not reflect instantly on the value of AKSEN shares. We've determined that the best lag for our case here is 2, since we believe the changes in PTF would apply 2 days later.

To determine the relevance of the lagged commodity values in forecasting the stock's market value, hypothesis tests, like the F-test or likelihood ratio test, can be carried out after the VAR model was estimated. The test's p-value determines if the market value of the stock is Granger-caused if it is less than a preset level of significance. This is generally set as 0.05.

We should note that the Granger causality does not indicate a direct correlation, but rather that one variable's ability to predict another, or in other words, if the changes in one reflect into the other.

Now that we've explained the model and our use, we will present the application in Python.

```python
import numpy as np
import matplotlib.pyplot as plt
import csv
import pandas as pd
from statsmodels.tsa.stattools import grangercausalitytests

df = pd.read_csv('new_aksen.csv', delimiter=';')
df = df.iloc[::-1]
df2 = pd.read_csv('ptf.csv', delimiter=';')
df2 = df2.iloc[::-1]
df3 = pd.read_csv('oil.csv', delimiter=';')
df3 = df3.iloc[::-1]

result = pd.merge(df2, df3, on=['Date'])
result = pd.merge(result, df, on=['Date'])
print()
cor = grangercausalitytests(
    result[['Value_x', 'Value_y']], 3)
print(cor)

fig, host = plt.subplots(figsize=(8, 5), layout='constrained')
ax1 = host.twinx()
ax2 = host.twinx()
ax1.plot(range(len(df['Date'])), df['Value'], color='red')
ax2.plot(range(len(df2['Date'])), df2['Value'])
plt.show()
```

Figure 15. Python codes for Granger Model

We've used an external Library that had both Granger model and VAR model integrated in. After that, we wrote our code, pulling the data from different Excel files, namely new_aksen.csv and ptf.csv. After pulling our data and fixing the bug that occurred in changing the Excel files from .xls to .csv, we've used the external codes to use the Granger tests related to our data. Our results after we ran the code is presented below:

Granger Causality

number of lags (no zero) 1

ssr based F test: F=30.0146 , p=0.0000 , df_denom=1054, df_num=1

ssr based chi2 test: chi2=30.1000 , p=0.0000 , df=1

likelihood ratio test: chi2=29.6794 , p=0.0000 , df=1

parameter F test: F=30.0146 , p=0.0000 , df_denom=1054, df_num=1

Granger Causality

number of lags (no zero) 2

ssr based F test: F=11.8273 , p=0.0000 , df_denom=1051, df_num=2

ssr based chi2 test: chi2=23.7671 , p=0.0000 , df=2

likelihood ratio test: chi2=23.5036 , p=0.0000 , df=2

parameter F test: F=11.8273 , p=0.0000 , df_denom=1051, df_num=2

Granger Causality

number of lags (no zero) 3

ssr based F test: F=6.5917 , p=0.0002 , df_denom=1048, df_num=3

ssr based chi2 test: chi2=19.9071 , p=0.0002 , df=3

likelihood ratio test: chi2=19.7216 , p=0.0002 , df=3

parameter F test: F=6.5917 , p=0.0002 , df_denom=1048, df_num=3

The underlined result is what we're looking for. The reasoning behind this is the other results are for 1 lagged causality and 3 lagged causality. This means we've eliminated results of 1 and 3 days lagged correlations. Above, we've explained why we are going with the 2 lagged causality model.

Granger Causality

number of lags (no zero) 2

ssr based F test: F=11.8273 , p=0.002 , df_denom=1051, df_num=2

ssr based chi2 test: chi2=23.7671 , p=0.002 , df=2

likelihood ratio test: chi2=23.5036 , p=0.002 , df=2

parameter F test: F=11.8273 , p=0.002 , df_denom=1051, df_num=2

Now we will explain the above graph and the underlined results from running the Python code. The F test, chi2 test and likelihood ratio tests are all hypothesis tests about how accurate our correlation runs.

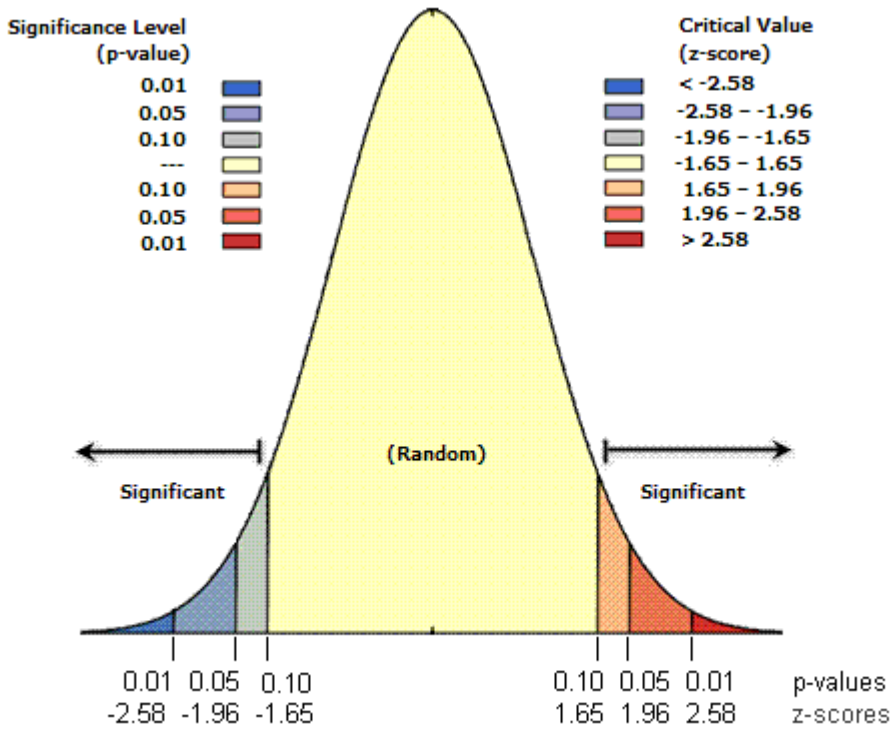

Figure 16. P-value Bell Curve

This is a bell curve, displaying the relevance of a correlation scale in the way of p-values. Our results bore a yield of 0.002 in p-value, which means our work will not be rejected, in other words random, and there is an significant correlation at hand. This could be understood beforehand after examining the graph of our results, shown below:

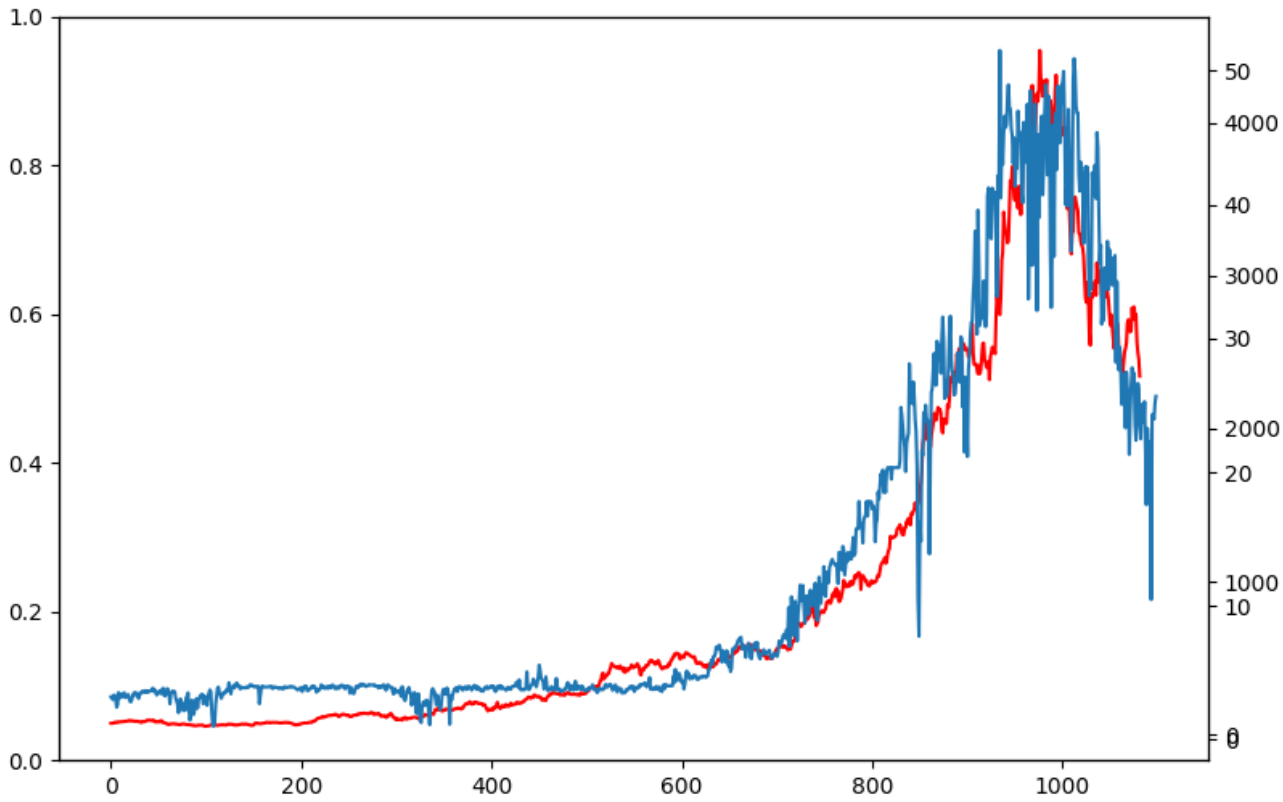

Figure 17. AKSEN compared to PTF

As evident from the graph above, there's a significantly similar trends of rise and fall in both PTF values (displayed in blue) and AKSEN share values (displayed in red).

In contrast to this, we've also done the same analysis based on Oil and AKSEN prices as a control constraint.

Granger Causality
number of lags (no zero) 2

ssr based F test: F=0.6114 , p=0.5428 , df_denom=1051, df_num=2

ssr based chi2 test: chi2=1.2287 , p=0.5410 , df=2

likelihood ratio test: chi2=1.2280 , p=0.5412 , df=2

parameter F test: F=0.6114 , p=0.5428 , df_denom=1051, df_num=2

Same analysis bore a result of p=0.5428 in comparison of Oil prices and AKSEN prices, in comparison to p=0.002 result of PTF and AKSEN prices, which shows a steep difference in the same number of lags, 2.

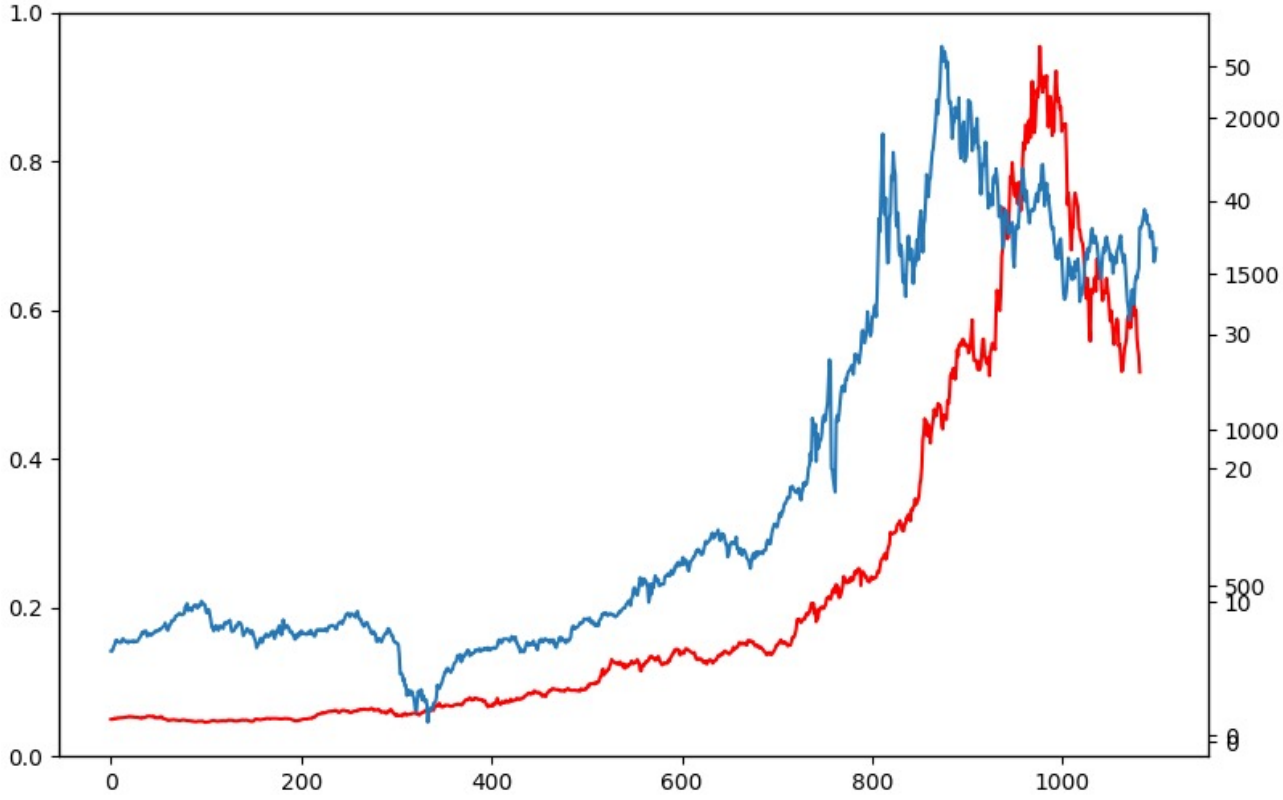

Figure 18. AKSEN compared to Brent Oil

Above is the Granger Causality graph of Oil prices versus AKSEN prices, dating from January 1st, 2019 to May 1st, 2023. When compared to the graph of PTF vs. AKSEN, we can see the steep lag difference between those graphs.

3.5. Discussion of results

Our results after using the Granger Causality Model and VAR Model bore healthy and understandable results. Our results showed a significant correlation between PTF prices and AKSEN prices, day by day compared from January 1st, 2019 to May 1st, 2023. We've also found that there's also a correlation between oil prices and AKSEN prices, although this correlation comes to fruition in a lagged manner compared to PTF price changes. Although we must not go over the fact that these changes over time were highly influenced by the economic state of Turkey, since high inflation rates and Turkish Lira's value declining over time causes the values of financial instruments to rise disregarding the external effects.

3.5.1 Theoretical

While doing analysis of the project, Excel and Python were used to search and analyze the financial data of the project. These tools were used by both the Management Engineering and Energy Systems Engineering students.

3.5.2 Practical

On the practical side of our project, we planned on influencing the investor's views on the energy sector and open a new window to energy sector investing with our findings.

3.6. Discussion of limitations

One of the most challenging parts of this project for us was finding and processing data on the energy sector. For example, EPİAŞ, or with it's other name, Energy Exchange İstanbul, keep some of their data and lessons confidential and do not share them. The fact that some data were not shared by some companies led to different sources in order to collect the data we wanted in the project. Also, learning different hypothesis methods and applying them to Python was a challenge since none of us are acquainted with the Python system on a daily basis. Stock market closing in the weekends and holidays and PTF being priced everyday was a constraint we had to work around, but we overcame it using VLOOKUP formula on Excel to only work the data with the same date.

• 4. INTEGRATION AND EVALUATION

○ 4.1. Integration

Management Engineers (ME) undertook the coordination of the project. In the research, technical knowledge and research on the side of the energy sector was done by Energy Systems Engineers (ESE). ESE students compiled the information they gathered from the sector. ME students took on the task of researching the financial or share-based side of the project and applying the research to come up with the result. The technical information gathered by the ESE students was reviewed by the ME students and thus a collaborative research was carried out. Apart from that, Energy System Engineers and Operations Engineers worked together at every stage.

- **5. SUMMARY AND CONCLUSION**

In this study, as Management Engineers and Energy Systems Engineers, we examined whether fluctuations in the energy market directly caused fluctuations in the stock exchange, in this case, the electricity index of Borsa İstanbul, with different methodologies and correlations to historic evidence of different commodities. We also explored how electricity is priced in Turkish markets and its different methodologies, mainly Market Clearing Price (PTF) and Merit Order. In all our works, EPİAŞ and TEİAŞ companies guided us in terms of legislation and pricing. After our research, we've came up with the result that the changes in both PTF and Brent Oil prices have a significant correlation in the changes of AKSEN shares.

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